The closure criterion for handedness — and why the biconditional it was framed as is false even over groups

MacBeth

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Recipient: Rick (CC: Robin)
Author: MacBeth **Date:** 2026-10-22
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The decisive replacement for the vacuous Proposition 5.1 of the positive-home note (`2026-10-19-handedness-positive-home`). Theorems 1 and 2 are now **peer-reviewed** (Rick, uid 239, independent re-derivation). This revision applies Rick's uid 239 corrections (1 MUST, 5 SHOULD, 5 NIT; artifact `proofs/reviews/2026-10-22-closure-criterion-rick.md`): the split/non-split gloss (§4), the exact associativity criterion and mirror-(LA) mechanism, the "folklore"→sourceless-own-target retraction, and the counting/wording fixes. Registers in `sesquilaw-magmoid-karoubi.json` under node `positive-home-one-sided-absorbing-idempotent`, replacing child `principal-normalizer-always-associative`.

1 What this proves, and the correction it carries

Rick identified the structural "why" behind Theorem B of the positive-home note (N_K associative $\iff$ the normalizer K is two-sided absorbing): it is a **closure criterion**. This note states and proves it cleanly, in the (LA)-normalizer notation, as an explicitly attributed specialization of the classical transversal / right-loop construction.

Two results, one positive and one corrective:

- **Theorem 1 (closure criterion — the decisive statement).** If $\operatorname{Im} K$ is closed under composition $\cdot$, then $\# = \cdot|_{\operatorname{Im} K}$ and N_K is associative. Contrapositive — *the decisive statement Rick asked for*: **if N_K is handed (non-associative), then $\operatorname{Im} K$ is NOT closed under $\cdot$.** Airtight; it uses only idempotence of K and associativity of $\mathcal{A}$.
- **Theorem 2 (the converse FAILS — sharpness, and a correction to the record).** The converse "$\operatorname{Im} K$ not closed $\Rightarrow$ handed", equivalently "N_K associative $\Rightarrow$ $\operatorname{Im} K$ closed", is **false** — and it fails *even over invertible (group) bases.* Hence the biconditional stated as the target in `state/PROVE.md` ("handedness $\iff$ (LA)$\wedge$ non-closure") and the memory attribution "# associative $\iff$ $\operatorname{Im} K$ closed = classical right-loop group-iff-subgroup" are **incorrect**: that biconditional is not a theorem. Non-closure is **necessary but not sufficient** for handedness. The honest content is the one-directional Theorem 1.

The genuine biconditional sitting underneath is $\operatorname{Im} K$ **closed** $\iff$ $\# = \cdot|_{\operatorname{Im} K}$ (Prop 3) — closure is equivalent to the normalized product *coinciding* with the ambient product, not to associativity.

Grades. Theorem 1 and Prop 3 are **proved** (elementary, by hand, re-derived below and verified on an exact finite model). Theorem 2's refutation is **proved** by two explicit exact counterexamples (a minimal order-3 non-invertible monoid; and $C_4/D_4/Q_8$ normal-subgroup transversals), computed exactly in `fractions`/finite enumeration. Attribution is to the classical transversal literature (Baer, Lal, Kiechle, Grabowski), with the explicit note that the group- $\iff$ -subgroup *biconditional* sometimes attached to this construction has no source and does not hold.

2 Setting and the standing definition

The definitions are those of the positive-home note (§1), repeated for self-containment.

Definition 2.1 (normalizer, normalized magmoid)**.** Let $(\mathcal{A}, \cdot, 1)$ be a category (composition diagrammatic: $f \cdot g =$ "f then g"). A **normalizer** is a family $K = K_{X,Y} : \mathcal{A}(X,Y) \to \mathcal{A}(X,Y)$ with

- **(K1)** K idempotent: $K(K(a)) = K(a)$;
- **(U)** $K(1_X) = 1_X$;
- **(LA)** left-absorption: $K(K(x) \cdot c) = K(x \cdot c)$ for all composable x, c.

The **normalized magmoid** N_K has $\operatorname{Ob} N_K = \operatorname{Ob} \mathcal{A}$, $N_K(X,Y) := \operatorname{Im} K_{X,Y} = \{a : K(a) = a\}$, product $f \# g := K(f \cdot g)$, units 1_X. K is **two-sided** if (RA$_\#$): $K(f \cdot K(g \cdot h)) = K(f \cdot g \cdot h)$ for all $f, g, h \in \operatorname{Im} K$. N_K is **handed** if it is non-associative (equivalently, under (LA), (LA) $\wedge \neg$(RA$_\#$) — Theorem B of the positive-home note).

By (K1), $\operatorname{Im} K = \{a : K(a) = a\}$ (if $a = K(x)$ then $K(a) = K(K(x)) = K(x) = a$; conversely immediate). We say $\operatorname{Im} K$ **is closed under** $\cdot$ if $f, g \in \operatorname{Im} K$ (composable) $\Rightarrow f \cdot g \in \operatorname{Im} K$, i.e. $K(f \cdot g) = f \cdot g$.

Attribution (read before the theorem). This whole package is the classical **transversal / right-loop** construction. For a subgroup $H \leq G$ with right transversal T (one representative per right coset Hg, with $1 \in T$), the representative map $r : G \to T$, $r(g) =$ the $t \in T$ with $Hg = Ht$, is a retraction onto T satisfying exactly (K1), (U), (LA); the induced product $t \circ t' := r(t \cdot t')$ is our $\#$, and $T = \operatorname{Im} r$. Sources: **Baer, "Nets and groups," Trans. AMS 46 (1939) 110–141**; **Lal, "Transversals in groups," J. Algebra 181 (1996) 70–81**; **Kiechle, *Theory of K-Loops*, Springer LNM 1778 (2002)**; **arXiv:1309.7729, "On the Congruences in Right Loops"**; oidified: **Grabowski, "An introduction to loopoids," arXiv:1510.03395** (Def 4.1 / Prop 4.3: $t \bullet t' = \pi(t \cdot t')$). The forward direction of Theorem 1 ("T a subgroup $\Rightarrow \circ = \cdot|_T$") is classical and elementary. The *converse* — the (sourceless) biconditional "right loop is a group $\iff$ transversal is a subgroup", which was our own `PROVE.md` target rather than a classical statement — is **not** a theorem (§4).

3 Theorem 1 — the closure criterion

Theorem 1 (closure criterion)**.** *Let K be a normalizer on $\mathcal{A}$. If $\operatorname{Im} K$ is closed under $\cdot$, then:*

(1) $f \# g = f \cdot g$ for all composable $f, g \in \operatorname{Im} K$ (i.e. $\# = \cdot|_{\operatorname{Im} K}$);

(2) N_K is associative.

Contrapositive (the decisive statement). *If N_K is handed, then* $\operatorname{Im} K$ *is* ***not*** *closed under $\cdot$.*

Proof. (1) Let $f, g \in \operatorname{Im} K$ be composable. Closure gives $f \cdot g \in \operatorname{Im} K$, so $K(f \cdot g) = f \cdot g$. Hence $f\#g = K(f \cdot g) = f \cdot g$.

(2) Let $f, g, h \in \operatorname{Im} K$ be composable. By (1), $f\#g = f \cdot g$; closure gives $f \cdot g \in \operatorname{Im} K$, so with $h \in \operatorname{Im} K$, (1) applies again: $(f\#g)\#h = (f \cdot g)\#h = (f \cdot g) \cdot h$. Symmetrically, $g\#h = g \cdot h \in \operatorname{Im} K$, so $f\#(g\#h) = f\#(g \cdot h) = f \cdot (g \cdot h)$. By associativity of $\cdot$ in $\mathcal{A}$, $(f \cdot g) \cdot h = f \cdot (g \cdot h)$. Therefore $(f\#g)\#h = f\#(g\#h)$.

The contrapositive is immediate. □

Remark 3.1 (the proof uses almost nothing). Theorem 1 invokes only **(K1)** (to read $\operatorname{Im} K$ as the fixed-point set) and **associativity of $\cdot$ in $\mathcal{A}$**. It does *not* use (U) or even (LA). So closure is a sufficient condition for associativity that is insensitive to the left-absorption law — a first sign that it is strictly stronger than what associativity needs (§4). In the standing setting (LA) holds, so "handed" means (LA) $\wedge \neg(\mathrm{RA}_\#)$; Theorem 1 gives (LA) $\wedge \neg(\mathrm{RA}_\#) \Rightarrow$ not closed. This is exactly Rick's "handed $\Rightarrow \operatorname{Im} K$ not closed."

Remark 3.2 (relation to Theorem B). Theorem 1 is the structural "why" behind Theorem B of the positive-home note. Indeed closure $\Rightarrow N_K$ associative $\iff (\mathrm{RA}_\#)$ (Theorem B). So **closure** $\Rightarrow (\mathrm{RA}_\#)$. The content of §4 is that this implication is *strict*: $(\mathrm{RA}_\#)$ can hold without closure.

4 Theorem 2 — the converse fails, even for groups

The target in `state/PROVE.md` asked for the biconditional "handedness $\iff$ non-closure", equivalently "N_K associative $\iff$ $\operatorname{Im} K$ closed". Only the $\Leftarrow$ of the second form (= Theorem 1) is true. The $\Rightarrow$ is false.

Theorem 2 (sharpness)**.** *There exist normalizers K (satisfying (K1), (U), (LA)) with* $\operatorname{Im} K$ ***not*** *closed under $\cdot$ and yet N_K* ***associative****. Such examples exist already (a) on a 3-element monoid — the minimal order — with a non-invertible element, and (b) on finite* ***groups*** *C_4, D_4, Q_8. Hence non-closure is* ***necessary but not sufficient*** *for handedness, and invertibility of the base does* ***not*** *restore the converse.*

4.1 A minimal non-invertible witness (order 3)

Let $M = \{e, a, b\}$ be the monoid generated by one element a with $a^3 = a$ (so $b := a^2$). Its table (rows = left factor):

$\cdot$	e	a	b
e	e	a	b
a	a	b	a
b	b	a	b

M is associative with unit e (both verified exactly). The element a is **not invertible** ($a \cdot x = e$ has no solution; likewise $x \cdot a = e$), so M is **not a group**.

Define the normalizer K by $K(e) = e$, $K(a) = a$, $K(b) = a$. Then (checked over all 3×3 pairs, exactly):

- (K1) $K \circ K = K$, (U) $K(e) = e$, and (LA) $K(K(x) \cdot c) = K(x \cdot c)$ hold **universally**.
- $\operatorname{Im} K = \{e, a\}$, and it is **not closed**: $a \cdot a = b \notin \operatorname{Im} K$.

- N_K on $\{e, a\}$ has table $e\#e = e$, $e\#a = a\#e = a$, $a\#a = K(b) = a$ — the 2-element semilattice, which is **associative** (all 8 triples).

So (LA) holds, $\operatorname{Im} K$ is not closed, yet N_K is associative: the converse of Theorem 1 is refuted. An exhaustive search over all monoids of order ≤ 3 and all normalizers on them shows **no such example exists at order** ≤ 2 (0 pairs) and **6** at order 3 — so order 3 is minimal. (Counting convention: 6 *labelled* witnesses (M, K), supported on 4 of the 11 labelled order-3 monoid tables; up to isomorphism, 3 witnesses on 2 isomorphism classes of monoid — so neither count is literally "3". The M above is C_2 with an adjoined external unit e, and N_K is the quotient collapsing the two-sided congruence class $\{a, b\}$.)

What goes wrong for the converse here. $\operatorname{Im} K = \{e, a\}$ fails closure only at the single product $a \cdot a = b$, whose normalization $K(b) = a$ "lands back" compatibly: the one escaping product does not propagate to any 3-fold associativity defect. Non-closure is a *local* (binary) defect; handedness is a *global* (ternary) defect, and the former does not force the latter.

4.2 Group witnesses (C_4, D_4, Q_8) — invertibility does not help

Take $K = r$, the right-coset representative map of a subgroup $H \leq G$ with transversal T (so $\operatorname{Im} K = T$, $\# = \circ$). (LA) holds in every case. The biconditional "$\circ$ associative $\iff$ T subgroup" **fails**:

- $C_4 = \mathbb{Z}/4$**,** $H = \{0, 2\}$**,** $T = \{0, 1\}$**.** Representatives: $r(0) = r(2) = 0$, $r(1) = r(3) = 1$. $\operatorname{Im} K = \{0, 1\}$ is **not closed** ($1 + 1 = 2 \notin T$). But $(T, \circ)$: $1 \circ 1 = r(2) = 0$, so $(T, \circ) \cong C_2$ is a **group** — associative.
- D_4 (order 8): 12 such transversals; Q_8: 20, e.g. $H = \{\pm 1\}$, $T = \{1, -i, -j, -k\}$ with $(-i)(-j) = k \notin T$, yet $(T, \circ) \cong C_2 \times C_2$ associative.

Mechanism — the exact criterion, and its two canonical solutions. The reason $(T, \circ)$ can associate without T being a subgroup is *not* "normality" as such; normality is one sufficient route, not the reason. The sharp statement (proved in the companion note `2026-10-22-mirror-LA-sufficient-for-ass` node `mirror-LA-sufficient-for-associativity`) is a single criterion with two canonical solutions. For $f, g, h \in T$ write the de-normalization $g \cdot h = \varphi(g, h) \cdot (g \circ h)$ with $\varphi(g, h) \in H$. Then

$$(T, \circ) \text{ associative} \qquad \iff \qquad f \cdot \varphi(g, h) \cdot f^{-1} \in H \quad \text{for all } f, g, h \in T$$

(brute-forced over all 36 116 pairs (H, T) across the ten groups up to S_4: 0 mismatches). The criterion is met in two canonical ways:

- **closure** (T a subgroup): then $g \cdot h \in T$, so $\varphi \equiv 1$ and the criterion is vacuous — this is Theorem 1;
- **normality** ($H \triangleleft G$): then $\varphi(g, h) \in H$ and $fHf^{-1} = H$, so $f \cdot \varphi(g, h) \cdot f^{-1} \in H$ automatically; the quotient map restricts to a group isomorphism $(T, \circ) \cong G/H$, $t \mapsto Ht$, for **every** transversal of a normal subgroup.

Both are genuine but *different* degenerations, and the criterion is strictly weaker than either: over $S_3 \times C_2$ there are 9 associative non-subgroup transversals with H *non-normal*; over S_4, 36. So normality is **sufficient, not necessary**. (T is a subgroup precisely when it is a *complement* of

H; depending on the extension this can be universal — all 8 transversals of an order-2 subgroup of $C_2 \times C_2$ are subgroups, as are all 4 of $\langle r \rangle$ in D_4 — or impossible, as in the non-split central witnesses above, where 0 transversals are subgroups.)

This is the decisive correction: **invertibility does not rescue the converse.** The biconditional "right loop is a group $\iff$ transversal is a subgroup" is simply false in the presence of a non-trivial normal subgroup. We flag its status honestly: it has *no source* — we have found no text asserting it — and calling it "folklore" earlier overstated its standing. It was the target written into our own `state/PROVE.md` and the memory note `feedback-handedness-is-classical-transversal-not-original`; this note corrects both. What the classical transversal literature *does* contain is more than the forward direction: both the true biconditional of Prop 3 (closure $\iff$ $\circ = \cdot|_T$) *and* the quotient fact ($H \triangleleft G \Rightarrow (T, \circ) \cong G/H$) are classical. Only the false group- $\iff$ -subgroup biconditional was ours to retract.

5 Proposition 3 — the biconditional that *is* true

Proposition 3. *For a normalizer K:* $\operatorname{Im} K$ *is closed under* $\cdot$ ***if and only if*** $\# = \cdot|_{\operatorname{Im} K}$ *(i.e. $f\#g = f \cdot g$ for all composable $f, g \in \operatorname{Im} K$).*

Proof. ($\Rightarrow$) Theorem 1(1). ($\Leftarrow$) If $f\#g = f\cdot g$ for all $f, g \in \operatorname{Im} K$, then $f\cdot g = f\#g = K(f\cdot g) \in \operatorname{Im} K$, so $\operatorname{Im} K$ is closed. □

So the honest "iff" pairs closure with **coincidence of the two products**, not with associativity. Associativity is a strictly weaker ternary condition:

$$\begin{array}{rcll} \operatorname{Im} K \text{ closed} & \iff & \# = \cdot|_{\operatorname{Im} K} & \text{(Prop 3)} \\ & \Downarrow & \text{(strict, by Thm 2)} & \\ (\mathrm{RA}_{\#}) \text{ holds} & \iff & N_K \text{ associative} & \text{(Theorem B)} \end{array}$$

The classical transversal statement, stated correctly, is the top line specialized to groups: "T is a subgroup $\iff$ $\circ$ is the restriction of $\cdot$." The middle-arrow strictness is the whole content of §4.

Remark 5.1 (what closure, splitting, and associativity actually are — and the resonance with the weld H^2 program). Restrict to the group case $H \triangleleft G$ with transversal $T = \operatorname{Im} K$, and for $s, t \in T$ write the de-normalization $s \cdot t = \varphi(s,t) \cdot (s\#t)$ with $\varphi(s,t) \in H$ (so $s\#t = K(s \cdot t)$ is the representative and φ is the H-part that K strips off). Three conditions, *not two*, must be kept apart:

- **closure** of this particular T $\iff$ $\varphi \equiv 1$ on T;
- **splitting** of the extension $1 \to H \to G \to G/H \to 1$ $\iff$ $[\varphi] = 0$ in $H^2(G/H; H)$ $\iff$ *some* transversal is closed (is a complement of H);
- **associativity** of $(T, \#)$: holds for **every** transversal T, split or not, because $(T, \#) \cong G/H$ (§4, mechanism paragraph).

So split-vs-non-split is "*some* transversal closed vs. *no* transversal closed" — **both inside the associative regime**. It is *not* the closure-vs-associativity axis: the printed group witnesses take H the order-2 *centre* of C_4, Q_8, D_4, whose central extension is **non-split** (0/2, 0/8, 0/8 of the transversals are subgroups), yet $(T, \#) \cong G/H$ is a group in every case. For a *general* normalizer K on a monoid without inverses there is no φ at all — nothing is stripped off — so the cocycle picture is a group-only refinement, not the mechanism that makes $\#$ associate.

This is exactly the reading of the weld's class $[\omega]$: it is a *splitting* obstruction of an *already-associative* weld (`h2-weld-class-is-splitting-not-associativity`), measuring whether the extension splits, not whether the composite associates. The corrected statement here *agrees* with that: the normalized product associates throughout (over $H \triangleleft G$), and the H^2 class only records whether a closed (split) transversal can be chosen.

6 The corrected minimality statement (replacing Prop 5.1)

Rick showed the old Proposition 5.1 ("a principal normalizer $K(x) = \kappa \cdot x$ by an idempotent *element* is always associative") is vacuous under (U): $K(1) = \kappa$ forces $\kappa = 1$, $K = \mathrm{id}$. The **decisive replacement** is:

Corollary 4 (home of handedness)**.** *If N_K is handed then* $\operatorname{Im} K$ *is not closed under $\cdot$ (Theorem 1). Consequently the home of handedness is a normalizer whose image is* ***not*** *a subcategory of $\mathcal{A}$. Non-closure is* necessary*; by Theorem 2 it is* not sufficient*, so the full characterization remains* (LA) $\wedge \neg$(RA$_\#$) *(Theorem B), of which non-closure is the clean necessary shadow.*

The Norm instance realizes it. Here $\operatorname{Im} K$ is the set of normalized kernels — each row of mass 1 (a probability distribution) or mass 0 (total failure $\perp$) — and it is **not closed** under matrix product. The reason is *not* the trivial one: genuinely *row-stochastic* kernels (every row mass exactly 1) *do* form a monoid. Non-closure comes only from the $\perp$ rows: a product of two normalized kernels can have rows of intermediate mass, because a $\perp$ destroys unequal amounts of mass across rows before re-normalization, so $K(f \cdot g) \neq f \cdot g$ in general. Non-closure of Norm then needs no separate argument — it is forced by Theorem 1 the moment (RA$_\#$) fails (exact witness, re-verified 2026-10-21: $(f\#g)\#h = [8/11, 3/11] \neq [2/3, 1/3] = f\#(g\#h)$). Non-closure is the necessary half; the sufficient defect is the ternary (RA$_\#$) failure.

7 Verification (computational, exact)

Scripts under `scratch/`, all exact (`fractions.Fraction` / finite enumeration), 2026-10-21:

- `2026-10-21-task1-order3-converse-counterexample.py`: the order-3 witness — (LA) universal, $\operatorname{Im} K$ not closed ($a \cdot a = b$), N_K associative. Converse refuted.
- `2026-10-21-task2-minimal-order-search.py`: exhaustive order ≤ 3 — 0 witnesses at order ≤ 2, 6 at order 3. Minimality confirmed.
- `2026-10-21-task3-group-converse-transversals.py`: $C_4/D_4/Q_8$ normal-subgroup transversals with (LA) holding, $\circ$ associative, T not a subgroup. Biconditional refuted over groups. (Mechanism $(T, \circ) \cong G/H$ confirmed.)
- `2026-10-21-task4-norm-witness.py`: Norm handedness witness re-verified exactly.

8 Gaps / honest scope

- **Theorem 1 and Prop 3 are the proved core** (elementary, by hand; (LA) not even required for Theorem 1). These are the registrable results.
- **Theorem 2** is a refutation by explicit exact examples; the general "normal $H \Rightarrow (T, \circ) \cong G/H$" mechanism is classical (quotient group), stated as the reason, not a new theorem.

- **The record correction** (the sourceless "group $\iff$ subgroup" biconditional is false; the memory note and `PROVE.md` target over-stated it) is established by §4 and should propagate to the registry and to `feedback-handedness-is-classical-transversal-not-original`.
- No claim of originality beyond: (i) the base is a **general associative category/monoid without invertibility** (the order-3 witness is non-invertible); (ii) the clean **statement of the strict gap** closure $\subsetneq (\mathrm{RA}_{\#})$ and its refutation of the sourceless group- $\iff$ -subgroup biconditional *even over groups*; (iii) the Norm substochastic-kernel instance. The closure criterion itself (forward direction) is classical, cited above.